

Question 1

In the context of Gradient Descent, what does the gradient vector $\nabla J(\theta)$ represent geometrically?

- A The direction of steepest descent (downhill)
- B The direction of steepest ascent (uphill) ✓ Correct Answer**
- C The lowest point of the function
- D The average rate of change

Question 2

Which of the following is the correct mathematical update rule for minimizing a cost function $J(\theta)$?

- A $\theta := \theta + \alpha \nabla J(\theta)$
- B $\theta := \theta - \alpha \nabla J(\theta)$ ✓ Correct Answer**
- C $\theta := \theta \times \alpha \nabla J(\theta)$
- D $\theta := \nabla J(\theta) - \alpha$

Question 3

What is the primary risk of setting the learning rate (α) too high?

- A The model will converge too slowly
- B The model will get stuck in a local minimum
- C The algorithm may overshoot the minimum and diverge** ✓ Correct Answer
- D The gradient will become zero immediately

Question 4

Why is "Vectorization" (using NumPy broadcasting) preferred over Python for loops for calculating distances in KNN?

- A It uses less RAM
- B It pushes the loop execution down to optimized C code** ✓ Correct Answer
- C It automatically selects the best value of K
- D It allows for variable-length arrays

Question 5

In KNN, what happens if Feature A has a range of [0, 1,000,000] and Feature B has a range of [0, 1] without normalization?

- A** The distance calculation will return an error
- B** Feature B will dominate the distance metric
- C** Feature A will dominate the distance metric ✓ Correct Answer
- D** The algorithm will automatically scale them

Question 6

Which distance metric corresponds to "Manhattan" geometry (grid-like movement)?

- A** L2 Norm (Euclidean)
- B** L1 Norm (Sum of absolute differences) ✓ Correct Answer
- C** Cosine Similarity
- D** Hamming Distance

Question 7

If a loss function $J(\theta)$ is "Convex," what does this imply about its optimization landscape?

- A It has multiple local minima (valleys)
- B It has ridges and plateaus that trap the optimizer
- C It has a single global minimum (bowl shape) ✓ Correct Answer**
- D It is impossible to minimize using Gradient Descent

Question 8

In the "Lazy Learning" approach of KNN, when does the heavy computation (distance calculation) occur?

- A During the training phase (.fit())
- B During the prediction phase (.predict()) ✓ Correct Answer**
- C When the data is loaded
- D During hyperparameter tuning

Question 9

 Incorrect

You are implementing Euclidean distance between matrix A (Test) and matrix B (Train) using the expansion $(A - B)^2 = A^2 + B^2 - 2AB$. Which NumPy operation corresponds to the $-2AB$ term?

A `np.sum(A) - np.sum(B)`

 Your Answer

B `A - B`

C `-2 * np.dot(A, B.T)`

 Correct Answer

D `-2 * (A * B)`

Question 10

What is the mathematical definition of a "Global Minimum"?

A A point where the gradient is zero

B The point with the lowest possible error across the entire domain

 Correct Answer

C A point lower than its immediate neighbors

D The point where the learning rate is zero